Can Satellites Replace the Field Plot? A Blind Benchmark of Satellite Forest Carbon and Conformal Crediting with Calibrated Risk

Kenny Tang
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
kenny.tang@binus.ac.id

Riki Awal Syahputra
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
riki.syahputra@binus.ac.id

Richard Hans
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
richard.hans001@binus.ac.id

Evelyn Jane Sutjiadi
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
evelyn.sutjiadi@binus.ac.id

Yenny Chen
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
yenny.chen@binus.edu

Jude Joseph Lamug Martinez
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia 11480
jmartinez@binus.edu

***Abstract*—Satellite platforms promise to replace field inventories in forest carbon crediting, but their estimates are rarely tested blind against the standard-formula answer. We separate the question into two isolated grids. The right grid computes sealed field ground truth to SNI 7724, IPCC, CDM, and Verra rules. The left grid sees only plot coordinates and dates and estimates carbon from time-matched Sentinel-1/2, Landsat, PALSAR, canopy height, and GEDI lidar. On 476 plots, blind error was 21% on 1 ha plots and 7–9% on project totals outside Indonesia, near 100% on small Indonesian plots. Counting trees on a 1 m canopy height map found 3% of the stems and 25% of the carbon (62% on the 2026 map). In 1,000 simulated projects the best map found a median 95% of the field carbon; satellite embeddings and a regional bias correction raised the share within $\pm 25\%$ from 80% to 88.5%, and on 35 unseen Bornean plots the frozen method found 85%. We propose conformal crediting, a lower bound calibrated on other projects with field truth. It met a 10% over-crediting target for five maps within 2.9 points, was invariant to map inflation, and credited 72–84% of the true stock without a field plot; the VM0047 deduction over-credited 23–29% of projects with field plots. No satellite estimate of annual growth beat assuming no change.**

***Index Terms*—forest carbon, MRV, remote sensing, cloud cover, conformal prediction, spatial cross-validation, carbon crediting, commit-reveal ledger, distributed processing**

I. Introduction

A forest carbon credit is issued on a number computed by a fixed chain. Field crews measure each stem's diameter D and height H and identify its species for wood density ρ; a pantropical allometric equation converts these to aboveground biomass (AGB) [1], and fixed factors convert AGB to carbon and CO_2 [2], [3]. The chain is the standard. The costly part is the field visit, above all in remote Indonesian forests. Satellite platforms, including our NusaCarbon prototype, promise to replace it, and global biomass maps exist [4], [5], but unverifiable estimates produce over-crediting [6].

How close is a satellite number to what the standard formula gives at the same place and time? Three traps inflate reported accuracy: *leakage*, when a model is trained or tuned on plots next to those it is scored on [7]; *temporal mismatch* between the field and image dates; and *scale*, when 0.05 ha plots are compared with 10 m to 4.6 km pixels.

We separate the problem into two grids (Fig. 1). The **right grid** is fixed: public field plots, the standard formula, a carbon density. The **left grid** is free, any satellite data and any algorithm, subject to one rule: it never sees the right grid's outcomes for the plots it is scored on. We ask: **RQ1** (methodology) how far a blind satellite estimate deviates from standard-formula ground truth, by plot size and aggregation; **RQ2** (AI) which inputs and learners close the gap, and whether trees can be counted from space; **RQ3** how often clouds hide an Indonesian plot, and whether uncertainty can be stated honestly at plot and project scale; **RQ4** whether a credit can be issued at a stated, verified risk of over-crediting, and whether growth can be estimated; and **RQ5** (distributed systems) whether extraction can be sharded and made fault tolerant with bit-identical output.

Contributions. (i) A blind, time-matched protocol with a commit-reveal ledger (476 stock and 196 repeat-census plots). (ii) A standards-based right grid with a Monte Carlo uncertainty budget. (iii) A plot-level clear-sky climatology for Indonesia. (iv) Missing-data conformal intervals under cloud. (v) *Conformal crediting*, a crediting rule with a calibrated over-crediting risk that is invariant to map inflation. (vi) A tree-by-tree satellite estimator and 1,000 simulated projects reported as carbon amounts. (vii) Fixes for each limit, frozen and tested blind on 35 unseen plots. (viii) A sharded, idempotent extraction runner.

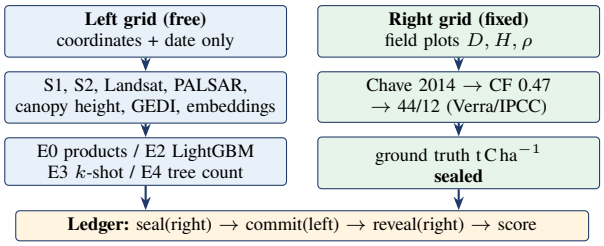


Fig. 1. Blind two-grid design. Only the manifest (plot id, coordinates, date, radius) crosses from right to left.

II. Related Work

Remote sensing. Optical and radar signals saturate over dense canopy. GEDI lidar samples height in 25 m footprints [9]; global maps include ESA CCI Biomass [4], Chloris [5], and canopy height [10], [11].

Validation and uncertainty. The BIOMASS package propagates the uncertainty of Chave's allometry [12]. Conformal prediction gives finite-sample intervals [8], adaptive with conformalized quantile regression (CQR) [13] and extended to missing covariates [14]. Conservative crediting asks that a credited stock exceed the truth with small probability [15], which Neeff solves parametrically [16].

III. Methodology

A. Right Grid: Ground Truth by the Standards

Data. The stock benchmark has 476 plots in 27 region blocks: 148 HCSA plots in six Indonesian provinces, nested 0.01/0.05 ha [17]; 142 GEO-TREES network plots, pantropical [18]; and 186 Labrière 1 ha units in French Guiana and Gabon [19]. For change, the SAFE census in Sabah gives 25×25 m plots measured stem by stem with day-precision dates from 2010 to 2020 [20]. We follow SNI 7724 [21] (five pools, precision ≤20%), the HCSA toolkit [22], IPCC [3] (Monte Carlo allowed), CDM AR-TOOL14 [23], Verra VM0047 [2] (deduct $U - 10\%$), and Gold Standard [24] (deduct the excess over 20%).

What a plot is. Coordinates only locate a plot; the ground truth is the list of trees measured inside it. A crew tags every stem above the minimum diameter and measures its diameter D at 1.3 m and height H; a botanist resolves its local name to species, whose wood density ρ (oven-dry mass per green volume) is looked up, or the genus or family mean. HCSA publishes the list: 2,528 stems of 370 species in 89 families, each with D, H, name, and the ρ of the ICRAF database [17]. GEO-TREES and Labrière publish plot biomass computed from their own lists; none of it reaches the left grid.

Tree, plot, and carbon. A tree's AGB follows Chave 2014 Eq. 4 [1],

$$\mathrm{AGB}_i = 0.0673\,(\rho_i D_i^2 H_i)^{0.976}\ [\mathrm{kg}], \qquad \sigma_{\ln} = 0.357, \quad (1)$$

with H measured (HCSA) or from a local $\ln H$–$\ln D$ model (Labrière) [12]. Tree biomass is expanded by the area in which a stem of its size is counted, and pools and carbon follow IPCC and SNI 7724:

$$\mathrm{AGB}_j = \sum_{i \in j} \frac{\mathrm{AGB}_i}{1000\,a(D_i)}, \quad a(D) = \begin{cases} 0.01, & 5 \le D < 15 \\ 0.05, & D \ge 15 \end{cases}$$
$$C_j = \mathrm{CF}\,(1+R)\,\mathrm{AGB}_j, \qquad \mathrm{CO_2e}_j = \tfrac{44}{12}\,C_j, \quad (2)$$

with CF = 0.47 and root-to-shoot $R = 0.37$ (SNI 7724, IPCC 2006) or 0.21 (IPCC 2019). The benchmark compares AGB, so CF and R cancel and every relative error holds for t C ha^{-1} and t CO_2e ha^{-1}. HCSA was recomputed tree by tree; plots with a missing input or an implausible stem (a *Gluta renghas* at $D = 480$ cm, a circumference typed as a diameter) were excluded, never imputed, and palms because Eq. 1 is for dicotyledonous trees.

Uncertainty and deduction. Plots are grouped into strata h with weights w_h, and the project uncertainty at 90% is $u = t_{0.95}\sqrt{\sum_h w_h^2 s_h^2/n_h}/\bar{b}$ (AR-TOOL14) [23]. AR-TOOL14 counts sampling error only, so we add IPCC Approach 2: 2,000 Monte Carlo draws per tree of diameter, height, wood density, allometric residual, and CF errors with the BIOMASS error models [12]. For change on the same plots, VM0047 v1.1 Eq. 28 gives $U = T\sqrt{SE_0^2 + SE_1^2 - 2\rho\,SE_0 SE_1}/\Delta C$, where ρ is the correlation of the two stock means; credits are multiplied by $1 - \min(1, \max(0, U - 0.1))$ [2].

B. Left Grid: Inputs and Estimators

For each plot, using only coordinates and the field date t_j, the left grid reads clear observations within $t_j \pm 90$ days: Sentinel-2 L2A bands and indices, Landsat 5–9, Sentinel-1 backscatter, the annual ALOS PALSAR mosaic, and elevation. Through Google Earth Engine it also reads the Meta 1 m and ETH 10 m canopy height maps [10], [11], GEDI L4A biomass and L2A height within 1 km [9], ESA CCI Biomass [4], and Hansen forest loss [25]; the fixes of Section IV-D add the AlphaEarth embedding (64 features per 10 m pixel) [26] and the 2026 canopy height map [27]. **E0** reads published products as they are. **E2** is LightGBM [28] on ln AGB, trained on all region blocks except the target and predicting it blind (leave-one-region-out, LORO), with smearing retransformation. **E3** recalibrates E2 by the ratio on k random plots of the target block. We score relative RMSE, $\mathrm{rRMSE} = 100\sqrt{(\hat{y}_j - y_j)^2}/\bar{y}$ for prediction $\hat{y}_j$ and field value y_j [9], [37], bias, R^2, and the error of the mean of n plots, which approximates a project total, with a region-block bootstrap (2,000 draws) and five seeds.

From pixels to carbon. A satellite records reflected light and radar echoes, not trees; Fig. 2 follows one plot from pixel to carbon. Chlorophyll absorbs red light and leaves reflect near-infrared, so NDVI [29], from Sentinel-2 bands B8 and B4, reads about 0.8 over canopy and 0.1–0.2 over bare soil.

E4: count the trees, then the formula. E4 rebuilds the right grid's tree list from the 1 m canopy height map, smoothed by a 1 m Gaussian. Tree tops are local maxima above 5 m in a window of radius $0.06H + 0.5$ m [30]; a crown grows from each top while the canopy stays above half its height; and the

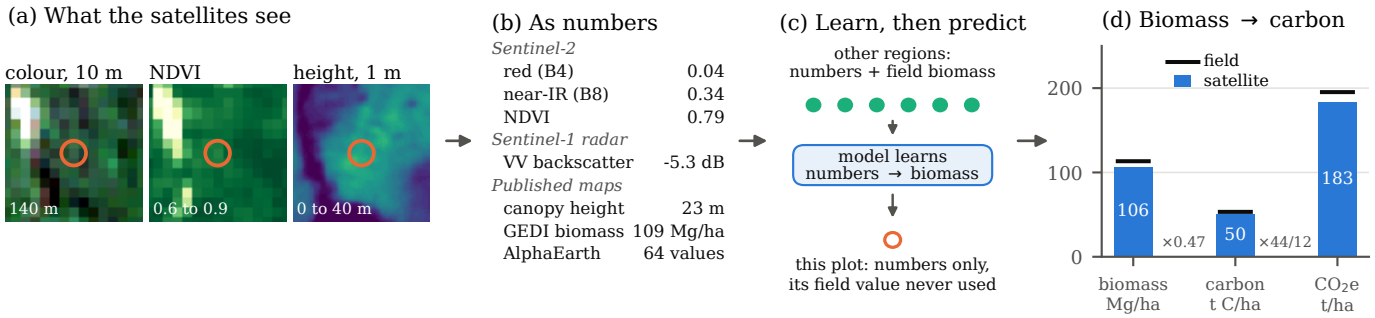


Fig. 2. From pixel to carbon on one Indonesian plot (HCSA Riau RU12, 0.05 ha). (a) The same 140 m around the field plot (circle), three ways: Sentinel-2 true colour and NDVI, each square one 10 m pixel (one clear day, May 2022), and the 1 m canopy height map. (b) Every satellite input for that footprint is a number: Sentinel-2 reflectance and NDVI, Sentinel-1 backscatter, published maps from other missions. (c) The model is fitted on other regions, where field biomass is known, and predicts this plot blind (LORO). (d) Fixed factors give carbon and CO_2e; here the satellite lands about 6% below the sealed field value.

stem diameter follows from height and crown diameter CD by the tropical-angiosperm model of Jucker *et al.* [31],

$$D = a\,(H \cdot CD)^b \text{ [cm]}, \qquad (a, b) = (0.486,\ 0.802), \quad (3)$$

for Indo-Malaya (realm-specific elsewhere). A satellite cannot identify species, so every tree takes the mean ρ of its region's tropical species in the Global Wood Density Database (0.574 for Southeast Asia) [12], and Eq. 1 gives its biomass. E4 uses no field data; *E4 scaled* multiplies it by the field-to-E4 ratio of the other regions.

C. Clouds

For 97 locations we read the Sentinel-2 scene classification of all 22,697 acquisitions of 2021–2023 in a 5×5 window of 20 m pixels (*clear* if every pixel is) and test the scene filter `CLOUDY_PIXEL_PERCENTAGE` < 30 against it. For a share q of target plots we remove the optical block and compare a naive model, modality dropout [32], a radar model, and imputation from radar.

D. Confidence: Plot and Project Intervals

All intervals target 90% and are calibrated on inner LORO residuals of the training regions only. CQR [13] scores $s = \max\{\hat{q}_{\text{lo}}(x) - y,\ y - \hat{q}_{\text{hi}}(x)\}$ on $y = \ln \text{AGB}$ and widens the quantile band by the conformal quantile of s. *Cloud-aware* CQR scores each calibration plot twice, as observed and with its optical block removed, and gives each test plot the threshold of its own missingness pattern, a two-pattern case of CP-MDA [14]. Its guarantee assumes missingness independent of the outcome, which clouds violate, so we measure coverage. A project's satellite mean has errors that share one regional bias, so its standard error is $SE^2 = s_w^2/n + s_b^2$, with s_w and s_b the within- and between-region spread of log residuals in other regions. With k field plots, the satellite becomes auxiliary data for a ratio estimator.

E. Conformal Crediting

A registry issues a number, not an interval. It should be a lower bound that exceeds the true stock in at most a share δ of projects [15], [16]. For a satellite map no design-based standard error exists, so we calibrate the bound on other projects with field truth. Project j has estimate $\hat{Y}_j$, true mean Y_j, and relative standard error r_j ($r_j = 1$ if none); its score is $z_j = \ln(\hat{Y}_j/Y_j)/r_j$. With scores from m calibration projects, a new project is credited

$$C = \hat{Y}\exp(-q\,r), \qquad q = z_{(\lceil (m+1)(1-\delta)\rceil)}. \quad (4)$$

If projects are exchangeable, $P(C > Y) \le \delta$, as in split conformal prediction with projects as units [8]; plots inside a project may be correlated in any way. The rule needs $m \ge 1/\delta - 1$ projects, nine for $\delta = 0.1$. If every map value is multiplied by c, every plain score shifts by $\ln c$, so q shifts by $\ln c$ and C is unchanged: *inflating a map earns no credit*. Regions of 16 or more plots are split into spatial sub-projects of about 10 plots (31 projects in 19 regions), each calibrated only on other regions. We score the realised over-crediting rate and the *yield*, the mean share C_j/Y_j of the true stock credited.

F. Ledger and Distributed Extraction

Every protocol step is an entry in an append-only log carrying the SHA-256 of the previous entry [33]: `seal`, `commit` (digest of each prediction file), `release`, and `reveal`, accepted only if the sealed file still matches; later commits are refused. Each extraction task is stored under its hash, so re-execution is idempotent. Transient errors are retried, and dense tiles are split and placed by consistent hashing [34] or the longest-processing-time (LPT) rule [35].

IV. Results

A. How Certain Is the Right Grid?

On the 148 HCSA plots, the Monte Carlo 95% interval of plot AGB has a median half-width of 31%. The standards' own choices are large: the SNI 7724 root-to-shoot ratio gives 13% more carbon than the 2019 Refinement, and the same Jambi data receive deductions of 24% (VM0047), 34% (AR-TOOL14), and 14% (Gold Standard), though AR-TOOL14 omits measurement and allometric error, 10–64% of the variance of the mean. For growth, VM0047 Eq. 28 with $\rho = 0$ deducts all credit for 13 of 15 SAFE blocks; with the measured ρ (0.45–0.99), seven keep some.

TABLE I
Blind rRMSE (%) Against the Right Grid, Leave-One-Region-Out

Left-grid estimate	All	GEO-T.	HCSA	1 ha
E0 Chloris, as published	49.2	37.4	107.1	30.7
E0 ESA CCI, as published	49.4	38.8	105.4	31.5
E0 GEDI L4A, as published	59.2	50.0	107.0	46.6
E2 (S1/S2, Landsat, PALSAR, DEM)	49.1	42.0	104.8	28.8
+ GEDI	45.5	36.6	102.4	24.5
+ canopy height	41.5	32.2	94.8	22.4
+ canopy height + GEDI	40.1	27.7	95.6	20.6
Gain over E2 (points)	−9.0	−14.3	−9.2	−8.2
Embeddings + CH + GEDI	**35.8**	**25.9**	**85.2**	**17.6**

475 plots with AGB $>1\,\mathrm{Mg\,ha^{-1}}$: GEO-TREES 142, HCSA 148 (0.05 ha), Labrière 185 (1 ha). Region-bootstrap 95% intervals of the gain row all exclude zero. Last row: Section IV-D, −4.3 points (−8.0 to −1.7) against canopy height + GEDI.

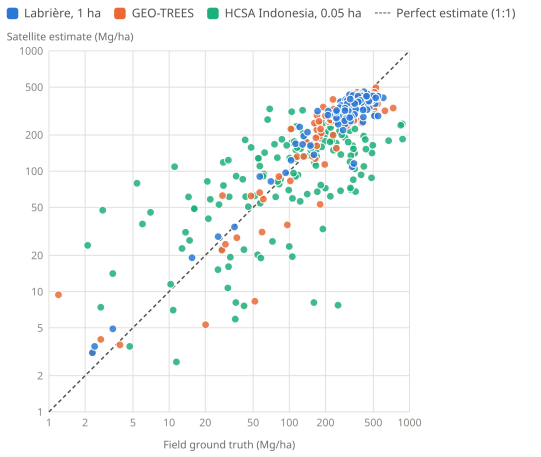


Fig. 3. Blind accuracy: E2 + canopy height + GEDI against the sealed field value, one dot per plot, leave-one-region-out. The 1 ha plots hug the 1:1 line; the 0.05 ha Indonesian plots scatter.

B. Blind Accuracy (RQ1) and Closing the Gap (RQ2)

Table I and Fig. 3 give the blind errors. The confirmatory E2, committed before the reveal, missed by 48.5% overall, 28.6% on the 1 ha units, 41.9% on GEO-TREES, and 102.9% on HCSA, where neither committed estimator beat predicting the mean. Plot size dominates the error. The largest gain came from inputs, not algorithms: adding the 1 m canopy height map and GEDI cut the overall error by 9.0 points (bootstrap 95% interval −13.0 to −6.1), to 20.6% on the 1 ha units and 27.7% on GEO-TREES, and raised R^2 from 0.29 to 0.53. Read as published, GEDI under-estimated by 29–41%: lidar is an input, not an answer.

Where the error comes from. Predictions saturate (above $400\,\mathrm{Mg\,ha^{-1}}$ the E2 mean is 293, below 50 it is 60), and an HCSA plot holds a median of 15 stems, so one large tree moves it by hundreds of $\mathrm{Mg\,ha^{-1}}$.

Aggregation and local plots. With canopy height and GEDI and no field plot, the median error of a block total was 7% on the 1 ha units, 9% on GEO-TREES, and 20% on HCSA. A ratio correction on $k = 10$–20 local plots (E3) brought E2 block totals from 29% to 9–10% on GEO-TREES and from 11% to 6% on Labrière, but hurt on HCSA until $k = 20$.

TABLE II
Simulated 1,000 ha Projects (6 of 10): Carbon Found, % of the Field

#	Region	Field, kt	Map	E4	E4 sc.	Credit
1	HCSA Riau	238	71	21	80	64
2	Labrière Nouragues	686	99	23	88	90
5	HCSA West Papua	254	91	46	180	83
7	Labrière Rabi	472	107	33	126	97
9	HCSA Jambi	287	93	23	88	86
10	Labrière Mondah	156	112	12	45	102
1,000 projects: median			95	23	85	86
5th–95th percentile			46–115	4–44	17–168	42–106
within ±25% of the field			80	0	40	76

Field: $\mathrm{kt\,CO_2e}$ from the measured trees, Eq. 1, $\times 0.47 \times \frac{44}{12}$. Map: E2 + canopy height + GEDI. E4 sc.: scaled on other regions. Credit: conformal, $\delta = 0.1$. HCSA projects: field 570–924 $\mathrm{stems\,ha^{-1}}$, E4 14–18.

C. Counting Trees from Space, and Ten Projects

What the field counts. The median HCSA plot holds 700 stems per hectare of five species. Stems of 5–15 cm are 75% of the count but 13% of the carbon; the 6% of 30 cm or more hold 58%. Giving every stem the regional default ρ leaves 91% of the total carbon, and 78–115% of a plot's.

What the satellite counts. On the HCSA plots E4 found a median of 20 trees per hectare: 3% of the stems, 11% of those of 15 cm or more. Without smoothing every stair step of the whole-metre map becomes a top (270 per hectare), so the count is a processing choice, not a measurement. E4 found 25% of the field carbon, 23–26% in every dataset. Scaled on other regions (×3.8), it found 97% of the total, but only 36% of plots fell within ±25% of the field value, against 60% for the map. Indices cannot do better: on 136 clear plots the median NDVI was 0.77 below $100\,\mathrm{Mg\,ha^{-1}}$ and 0.83 above 300 (Fig. 2).

Ten projects. Table II reports the carbon each grid finds in a simulated 1,000 ha project of 10 random plots of one region (6 of 10 shown; all rows in the extended version).

D. Toward Production: Fixes and an Unseen Test

We attacked each limit after the reveal (exploratory, Table III), froze the method, and tested it blind (Fig. 4). **Inputs.** The embedding cut plot error from 40.1 to 35.8% (Table I); the 2026 map added little. **Bias.** The map's bias repeats within a realm, so we multiply it by $e^{-\mu}$, μ the mean log ratio of map to field over the realm's other regions; HCSA projects within ±25% rose from 56 to 68%. **Credit.** Errors are shared within a region, so the region is the exchangeable unit: one score per region in Eq. 4 cut over-crediting from 13.6 to 9.7%. **Frozen method.** Per 1 ha cell: Earth Engine features, the LightGBM map, the estimate $\hat{Y}e^{-\mu}$, the credit $\hat{Y}e^{-q}$ (q over field-verified regions), and growth from the IPCC default [3]. **Unseen test.** 35 plots of 0.28 ha measured in 2020 in Danum, Sabah [36] (4,201 live stems, $366\,\mathrm{Mg\,ha^{-1}}$), were sealed before the frozen method, trained without the three GEO-TREES Danum plots, committed. The corrected estimate found 85% of the field carbon (93% of 10-plot draws within ±25%) and the credit 59%, never above the field.

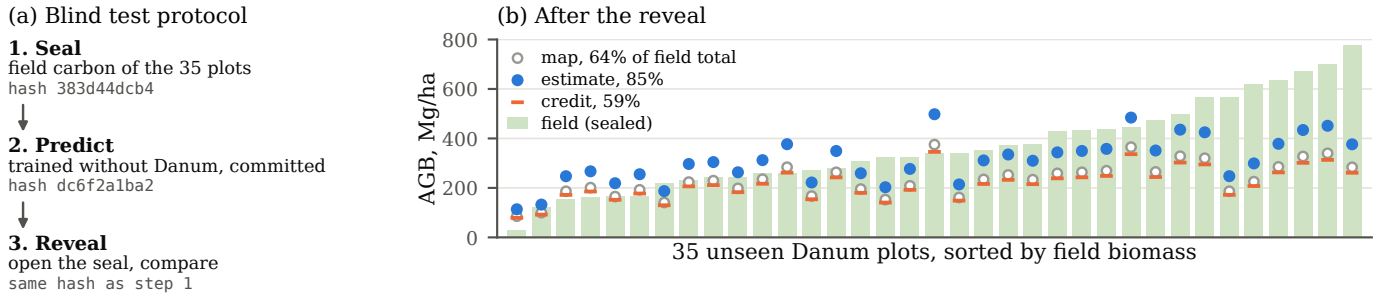


Fig. 4. The blind test on unseen plots in Danum, Sabah. (a) The field carbon was sealed by hash, the frozen method's predictions committed, then the seal opened. (b) Each plot: field biomass, the LightGBM map, the estimate (map $\times e^{-\mu}$) and the credit (map $\times e^{-q}$); totals as % of the field.

TABLE III
Fixes and the Unseen Danum Test, % of the Field Carbon

	1,000 projects			Danum
Estimate	**Median**	**5–95th**	**±25%**	**total**
Map of Table II	95	46–115	80.0	–
Embeddings + CH + GEDI	95	61–111	84.2	64
$\times e^{-\mu}$, realm bias	101	78–132	**88.5**	**85**
credit, region-level q	87	56–104	79.4	59
Jucker 2018 on CHMv2, unfitted	101	41–196	62.5	94

Credit over the field: 9.7% of projects. Jucker on Danum: post-reveal.

E. Clouds and Confidence (RQ3)

How often is a plot clear? At a median Indonesian location only 21.5% of Sentinel-2 acquisitions see the plot clear (Americas 39%), but revisit compensates: the median gap between clear looks is 10 days. *The scene filter is the real problem*: in Indonesia it keeps 11.5% of acquisitions, 34% of them cloudy over the plot, and discards 65% of the clear ones. Removing every optical feature raised the error only from 48.0% to 50.0%.

Which intervals are honest? CQR is the only interval method that reaches the 90% target with usable width (89.9%, upper bound 5.3 times the lower). Normalized conformal under-covers (79.4%). When every target plot is clouded, plain CQR falls to 86.5%, and cloud-aware CQR keeps 90.5% by widening honestly (+4.0 points, interval +1.9 to +6.6). HCSA stays under-covered (68–81%). A metadata confidence score like our platform's carries no information about error (partial rank correlation −0.03 to +0.07).

Projects. For 19 projects, an interval with independent plot errors covered only 74% (HCSA 33%); the two-component interval covered all 19. As auxiliary data, the satellite cut the field plots needed for ±10% by about half outside Indonesia.

F. From Intervals to Credits and Growth (RQ4)

We credit the 31 projects at $\delta = 0.10$. Crediting the raw map over-credits 35.5% of projects; VM0047 on the two-component uncertainty credits only 48% of the true stock. Conformal crediting meets its target: 9.7% of projects are over-credited, and 72% of the true stock is credited without a field plot in the project, 84% with the canopy height and GEDI map. A better map buys credits; it does not buy risk. With 5–10 field plots and a ratio estimator, VM0047 still over-credits 23–29% of projects, because its deduction starts only above 10% uncertainty; the design lower bound and studentized conformal meet the target (9.9 and 9.6%).

Whatever the map? On five maps, including E2 inflated by 20%, over-crediting stayed within 2.9 points of the target.

Growth. Excluding oil palm, 196 SAFE plots gained a mean 1.87 Mg ha^{-1} yr^{-1}. The Chloris difference did no better than predicting no change (RMSE 8.06 vs 8.12): a stock error of 30–100% cannot resolve growth of 1% per year. The mean rate of the other blocks did best (block MAE 0.98 vs 2.05).

G. Systems (RQ5)

The extraction ran 2,052 tasks on 24 workers in 2.1 h (worst-worker load 6.75 to 1.40 after splitting dense tiles), bit-identical to one worker under faults and resumes.

V. Discussion

Can satellites replace the field plot? Not at plot scale, and not for growth. Open satellite data put a 1 ha plot within about 21% (airborne lidar 13% [37], radar 31–38% [38]), a 0.05 ha plot within about 100%. Nor can they count trees: the 1 m map shows a canopy surface, not stems, and the field formula applied to it finds a quarter of the carbon, 62% on the 2026 map (4 m taller). For growth, no satellite estimator beat a regional average. What satellites do well is *replace part of the inventory*: project totals, and half the field plots where plots are large.

The ground truth is a model too. With valid standards 13% apart in carbon and 14–34% apart in deductions, "100% accuracy" has no meaning.

Confidence must be calibrated, not graded. The $U - 10\%$ deduction is not a lower confidence bound. Conformal crediting makes the over-crediting rate a design parameter and removes the reward for inflating the map; the target is Grassi's and Neeff's, the machinery split conformal prediction [8], [39].

Roadmap. NusaCarbon should adopt the frozen method of Section IV-D, plot-level cloud masking, and cloud-aware CQR; stems remain unseen, which motivates drone lidar.

Threats to validity. Only E0, E2, the first change estimates, and the Danum test are confirmatory; all other experiments were defined after the reveal and are exploratory. Danum is one valley of tall forest. GEDI and Chloris may have been

calibrated on some of our sites. HCSA and Labrière dates are known only to the year. Projects are region blocks, not registered projects.

VI. Conclusion

We tested blind, against a sealed standard-formula right grid, whether open satellite data can replace the field plot. Satellites reached 21% error on 1 ha plots and 7–9% on project totals, but failed on small plots, tree counts, and growth. Embeddings and a realm bias correction put 88.5% of simulated projects within $\pm 25\%$ of the field carbon, and the frozen method found 85% on unseen Bornean plots. Plot-level cloud masking nearly triples usable optical data, and conformal crediting issues credits at a stated risk without a field plot, unaffected by map inflation.

Data and Code Availability

Field data are public [17]–[20]. Code, results, an extended version with all tables, and an interactive dashboard are at https://github.com/kennytangg/carbon-paper.

References

[1] J. Chave *et al.*, "Improved allometric models to estimate the aboveground biomass of tropical trees," *Global Change Biol.*, vol. 20, no. 10, pp. 3177–3190, 2014, doi: 10.1111/gcb.12629.

[2] Verra, *VM0047 Afforestation, Reforestation, and Revegetation*, ver. 1.1, May 2025. [Online]. Available: https://verra.org/wp-content/uploads/2025/05/VM0047-v1.1_Final.pdf

[3] IPCC, *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 4*, Ch. 4 Forest Land, Tables 4.4 and 4.9 (Updated). Geneva: IPCC, 2019.

[4] M. Santoro *et al.*, "The global forest above-ground biomass pool for 2010 estimated from high-resolution satellite observations," *Earth Syst. Sci. Data*, vol. 13, pp. 3927–3950, 2021.

[5] Chloris Geospatial, "Global aboveground biomass stock and change, 2003–2019," distributed by the Microsoft Planetary Computer. [Online]. Available: https://planetarycomputer.microsoft.com/dataset/chloris-biomass

[6] T. A. P. West *et al.*, "Action needed to make carbon offsets from forest conservation work for climate change mitigation," *Science*, vol. 381, no. 6660, pp. 873–877, 2023.

[7] P. Ploton *et al.*, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nat. Commun.*, vol. 11, Art. no. 4540, 2020.

[8] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *J. Amer. Statist. Assoc.*, vol. 113, no. 523, pp. 1094–1111, 2018, doi: 10.1080/01621459.2017.1307116.

[9] L. Duncanson *et al.*, "Aboveground biomass density models for NASA's Global Ecosystem Dynamics Investigation (GEDI) lidar mission," *Remote Sens. Environ.*, vol. 270, Art. no. 112845, 2022.

[10] J. Tolan *et al.*, "Very high resolution canopy height maps from RGB imagery using self-supervised vision transformer and convolutional decoder trained on aerial lidar," *Remote Sens. Environ.*, vol. 300, art. 113888, 2024, doi: 10.1016/j.rse.2023.113888.

[11] N. Lang, W. Jetz, K. Schindler, and J. D. Wegner, "A high-resolution canopy height model of the Earth," *Nature Ecol. Evol.*, vol. 7, pp. 1778–1789, 2023, doi: 10.1038/s41559-023-02206-6.

[12] M. Réjou-Méchain, A. Tanguy, C. Piponiot, J. Chave, and B. Hérault, "BIOMASS: An R package for estimating above-ground biomass and its uncertainty in tropical forests," *Methods Ecol. Evol.*, vol. 8, no. 9, pp. 1163–1167, 2017.

[13] Y. Romano, E. Patterson, and E. Candès, "Conformalized quantile regression," in *Adv. Neural Inf. Process. Syst. 32*, 2019, pp. 3543–3553.

[14] M. Zaffran, A. Dieuleveut, J. Josse, and Y. Romano, "Conformal prediction with missing values," in *Proc. 40th Int. Conf. Machine Learning (ICML)*, PMLR vol. 202, 2023, pp. 40578–40604.

[15] G. Grassi *et al.*, "Applying the conservativeness principle to REDD to deal with the uncertainties of the estimates," *Environ. Res. Lett.*, vol. 3, art. 035005, 2008, doi: 10.1088/1748-9326/3/3/035005.

[16] T. Neeff, "What is the risk of overestimating emission reductions from forests – and what can be done about it?" *Climatic Change*, vol. 166, art. 26, 2021, doi: 10.1007/s10584-021-03079-z.

[17] High Carbon Stock Approach, "Forest field plot data for indicative HCS mapping, Indonesia," Fair Forward/GIZ and JKPP, 2023, CC BY-SA 4.0. [Online]. Available: https://huggingface.co/datasets/fair-forward/hcsa_indonesia

[18] GEO-TREES, "Biomass reference plot data," IIASA, accessed Oct. 2026. [Online]. Available: https://geo-trees.nodes.iiasa.ac.at/page/data

[19] N. Labrière *et al.*, "In situ reference datasets from the TropiSAR and AfriSAR campaigns in support of upcoming spaceborne biomass missions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3617–3627, 2018, doi: 10.1109/JSTARS.2018.2851606.

[20] M. Svátek *et al.*, "Tree census data from the SAFE Project 2011–2020," Zenodo, 2025, doi: 10.5281/zenodo.14882506.

[21] Badan Standardisasi Nasional, *SNI 7724:2011 Pengukuran dan penghitungan cadangan karbon – Pengukuran lapangan untuk penaksiran cadangan karbon hutan (ground based forest carbon accounting)*. Jakarta: BSN, 2011.

[22] G. Rosoman, S. S. Sheun, C. Opal, P. Anderson, and R. Trapshah, Eds., *The HCS Approach Toolkit*, ver. 2.0, Module 4. Singapore: HCS Approach Steering Group, 2017.

[23] UNFCCC CDM Executive Board, *A/R Methodological Tool 14: Estimation of Carbon Stocks and Change in Carbon Stocks of Trees and Shrubs in A/R CDM Project Activities*, ver. 04.2.

[24] Gold Standard, *Afforestation/Reforestation GHG Emissions Reduction & Sequestration Methodology*, ver. 2.1, May 2024.

[25] M. C. Hansen *et al.*, "High-resolution global maps of 21st-century forest cover change," *Science*, vol. 342, pp. 850–853, 2013, doi: 10.1126/science.1244693.

[26] C. F. Brown *et al.*, "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data," arXiv:2507.22291, 2025.

[27] J. Brandt *et al.*, "CHMv2: Improvements in global canopy height mapping using DINOv3," arXiv:2603.06382, 2026.

[28] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Adv. Neural Inf. Process. Syst. 30*, 2017, pp. 3146–3154.

[29] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, 1979, doi: 10.1016/0034-4257(79)90013-0.

[30] S. C. Popescu and R. H. Wynne, "Seeing the trees in the forest: Using lidar and multispectral data fusion with local filtering and variable window size for estimating tree height," *Photogramm. Eng. Remote Sens.*, vol. 70, no. 5, pp. 589–604, 2004.

[31] T. Jucker *et al.*, "Allometric equations for integrating remote sensing imagery into forest monitoring programmes," *Global Change Biol.*, vol. 23, no. 1, pp. 177–190, 2017, doi: 10.1111/gcb.13388.

[32] N. Neverova, C. Wolf, G. Taylor, and F. Nebout, "ModDrop: Adaptive multi-modal gesture recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2016, doi: 10.1109/TPAMI.2015.2461544.

[33] S. Haber and W. S. Stornetta, "How to time-stamp a digital document," *J. Cryptology*, vol. 3, no. 2, pp. 99–111, 1991.

[34] D. Karger, E. Lehman, T. Leighton, R. Panigrahy, M. Levine, and D. Lewin, "Consistent hashing and random trees," in *Proc. 29th ACM Symp. Theory Comput.*, 1997, pp. 654–663.

[35] R. L. Graham, "Bounds on multiprocessing timing anomalies," *SIAM J. Appl. Math.*, vol. 17, no. 2, pp. 416–429, 1969.

[36] N. Keller *et al.*, "'Danum Revised' forest field plot data for thirty-five 0.28 hectare field plots, measured in 2020," Zenodo, 2023, doi: 10.5281/zenodo.8042468.

[37] T. Jucker *et al.*, "Estimating aboveground carbon density and its uncertainty in Borneo's structurally complex tropical forests using airborne laser scanning," *Biogeosciences*, vol. 15, pp. 3811–3830, 2018, doi: 10.5194/bg-15-3811-2018.

[38] A. Berninger, S. Lohberger, M. Stängel, and F. Siegert, "SAR-based estimation of above-ground biomass and its changes in tropical forests of Kalimantan using L- and C-band," *Remote Sens.*, vol. 10, no. 6, art. 831, 2018, doi: 10.3390/rs10060831.

[39] L.-C. Zhang and T. Tuoto, "Conformal confidence intervals with an application to small area estimation," arXiv:2608.02766, 2026.